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Bernstein Energy & Power: Heat pumps - an underappreciated lever of power demand growth?

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BERNSTEIN ENERGY & POWER: HEAT PUMPS - AN UNDERAPPRECIATED LEVER OF POWER DEMAND GROWTH IN EUROPE?

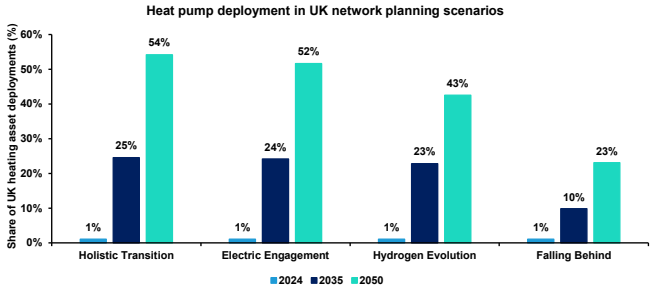
Written by Deepa Venkateswaran and Rory Graham-Watson

The 2022 European energy crisis and the 2026 Iranian war have spurred significant adoption of decarbonisation technologies in Europe, including increased uptake of EVs and household solar/ battery installations. In this Blast we analyse the prospects for an acceleration of the domestic heat pump roll-out in the wake of the latest crisis, as well as the potential beneficiaries in this scenario.

INTRODUCTION

As set out in Exhibit 1, the heat pump is a core technology in the decarbonization plans of European national governments, including the UK. The UK Government is targeting over 450,000 new installations per year by 2030, with the National Energy System Operator contemplating 10-25% of households being fitted with a device by 2035, vs ~1% currently. Additionally, under the Labour government's Future Homes Standard, all new-build houses in the UK from 2028 onward must be fitted with a heat pump. In the EU, the 2022 REPowerEU Plan contemplates the installation of 30m heat pumps by 2030, whilst the 2026 Energy Plan/AccelerateEU contemplates 4m heat pump sales per year by 2030. In this edition of the *Blast* we set out an overview of heat pump technology including its history, mechanisms and key variants, as well as examining its potential implications for the broader electricity system.

EXHIBIT 1: **Heat pump adoption is a key pillar of the UK's decarbonisation plans**



Source: UK National Energy System Operator, Bernstein analysis

WHAT IS A HEAT PUMP?

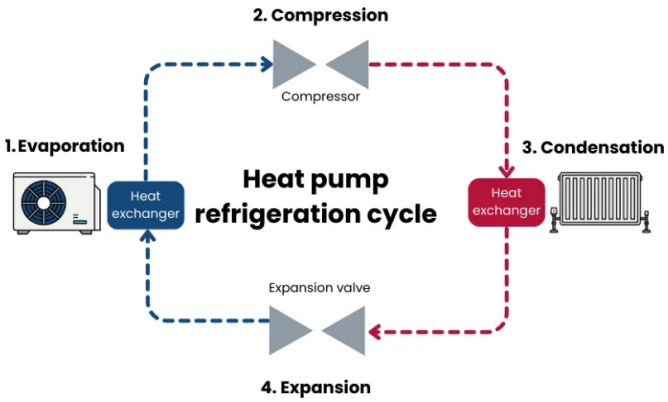
A heat pump is an electrically-powered heating device that uses a combination of ambient heat and compression technology to heat a building or other structure. The most commonly deployed version of the technology, the air source heat pump (ASHP), absorbs ambient heat from the air. Other versions, e.g. ground source heat pumps (GSHP) source heat from underground pipe networks. Heat pumps can enjoy significant energy efficiency benefits as the device is able to transfer more heat energy into the heated building than the electrical energy it uses in its operation.

HOW DO HEAT PUMPS WORK?

Heat pumps work on the basis of the 4-step refrigeration cycle, i.e. 1) Evaporation, 2) Compression, 3) Condensation and 4) Expansion.

See the Disclosure Appendix of this report for required disclosures, analyst certifications and other important information. Alternatively, visit our Global Research Disclosure Website.
First Published: 05 Jun 2026 04:55 UTC Completion Date: 04 Jun 2026 18:40 UTC

EXHIBIT 2: **Illustrative air source heat pump operating cycle**



Source: Energy Saving Trust

Evaporation

In air source heat pumps, fans are used to blow ambient outdoor air over a heat exchanger containing a chemical refrigerant in a liquid form. This liquid chemical refrigerant has a sufficiently low boiling point that even the ambient outdoor heat is sufficient to induce its evaporation into a low-pressure, low-temperature gas.

Compression

The newly-evaporated gas is directed into an electrically-powered compressor. The compression process further raises the temperature of the gas by raising its pressure.

Condensation

Following the compression process, the gas is transferred to a second heat exchanger which transfers its stored heat energy into a water circuit, thereby warming the water which subsequently either powers the heating system emitters (e.g. radiators or underfloor heating installations) or is used for the hot water supply.

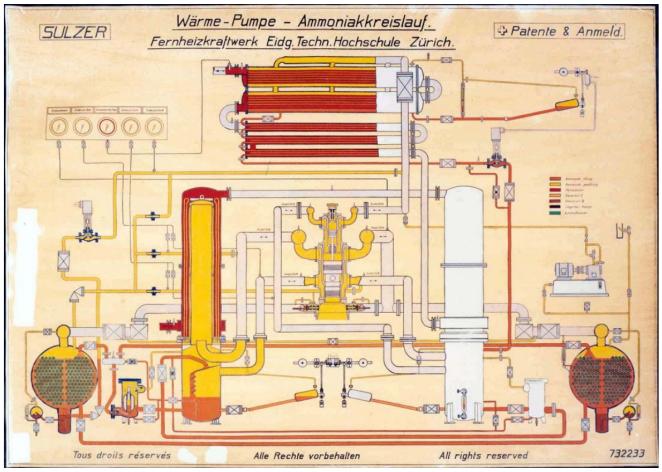
Expansion

Having transferred its heat to the hot water system, the newly-cooled but still compressed gas is passed through an expansion valve to lower its pressure. The decompression process further lowers the temperature of the gas such that it is again ready to absorb ambient heat from the outdoor surroundings.

HISTORY OF HEAT PUMP TECHNOLOGY

The first instance of heat pump technology in Europe can be traced to Switzerland. Reportedly "as early as 1852, Lord Kelvin had an intuition regarding the heat pump, in remarking that a 'reverse heat engine' could be used not only for cooling but also for heating.' The first large scale heat pump was installed in the US in 1948 before the 1970s oil crisis accelerated the momentum of the technology.

EXHIBIT 3: **A 1.86 MW Sulzer heat pump with 3 three-stage piston compressors, 1942**



Source: Martin Zogg, European Heat Pump Association

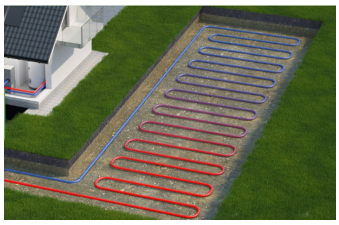
TYPES OF HEAT PUMP

The two primary types of heat pump are 1) air source and 2) ground source. The former draws ambient heat from the surrounding air, whilst the latter is buried underground and draws ambient heat from the surrounding earth. Air source heat pumps are significantly more prevalent because 1) they require less space and 2) are significantly less costly to install, although they are also somewhat less efficient to operate because of variations in air temperature, which varies more than ground temperature.

EXHIBIT 4: **An air source heat pump**

Source: EDF

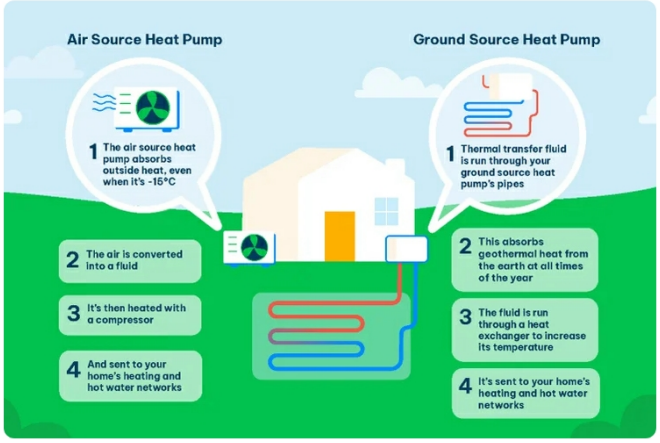
EXHIBIT 5: **A ground source heat pump**



Source: Energy Saving Trust

We set out a comparison of the operating mechanisms of air source and ground source heat pumps in Exhibit 6 below.

EXHIBIT 6: **Air source and ground source heat pump operating mechanisms**

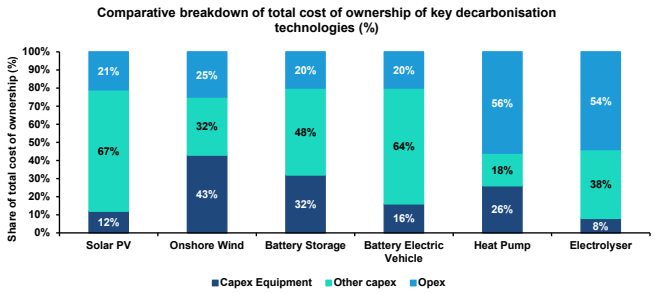


Source: The Eco Experts

ILLUSTRATIVE ECONOMICS

Heat pumps are an alternative heating technology to traditional mechanisms such as oil- or gas-fired boilers. The competitiveness of heat pumps vs alternatives is a function of 1) the upfront cost (in turn primarily a function of site-specific factors) and 2) the ongoing operating cost (a function of e.g. the quality of installation and the ratio of electricity to gas prices in the relevant market). Unlike other decarbonization technologies, the total costs of ownership of heat pumps have an opex rather than a capex skew - see Exhibit 7.

EXHIBIT 7: **Illustrative split of total cost of ownership of key decarbonisation technologies**



Values estimated from Figure 7.17 of IEA Energy Technology Perspectives 2026
Source: IEA*, Bernstein analysis

Installation costs

Heat pumps are typically significantly more expensive to install than comparable gas boilers on account of the significantly greater complexity of the process, which may require upgrades to pipework and the installation of additional radiators as well as upgrades to the existing hot water cylinder.

Operating costs

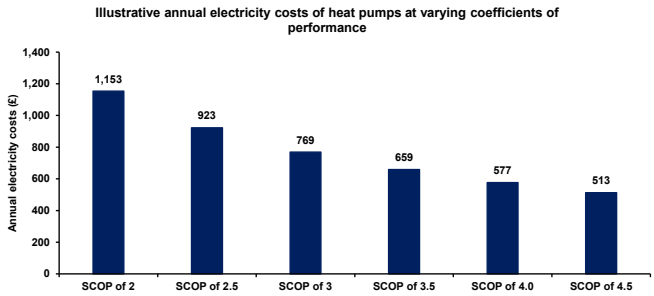
Excluding initial installation costs, the key considerations driving the technology decision between a gas boiler and an electric heat pump are 1) the comparative efficiency ratings and 2) the comparative fuel costs. Where correctly installed, heat pumps are able to achieve an efficiency rating of up to ~400%, i.e. using only 25% of the heat energy they generate as electrical energy to power the heating process. This is achieved by the physics of the evaporation, compression, condensation and expansion processes. By contrast, a gas boiler would typically achieve a maximum efficiency rating of ~93%, using more gas input than heat output generated. However, whilst heat pumps are considerably more efficient than gas boilers, the electricity that they use for fuel is also significantly more expensive in several countries including the UK.

The ratio of heat output to the electric energy input is known as the Coefficient of Performance ('COP'). Alongside the cost of electricity and the temperature differential between outside and inside, the COP is a key determinant of the operating costs of a heat pump.

The operating efficiency of a heat pump varies with: 1) the external temperature, with greater efficiency on warmer days; and 2) the internal temperature, with greater efficiency when the internal heating and hot water systems run at a lower temperature; alongside other factors pertaining to the system design, as we explain in more detail later in this *Blast*.

We set out in Exhibit 8 the indicative annual electricity costs for a heat pump generating ~9,000 kWh of heat energy per year and operating at varying efficiency levels. This analysis assumes electricity costs at 26.11p/kWh, in line with the recently published UK price cap rates for Q3 2026.

EXHIBIT 8: **Achieved operating efficiency is a key driver of the total cost of ownership for a heat pump**



Source: Bernstein analysis and estimates

In a domestic context, a homeowner would typically consider installing an air source heat pump as an alternative to a gas boiler. As we set out in Exhibit 9, along with the prevailing level of electricity prices, the operating efficiency is a key variable in this decision, with a SCOP of higher than 3 typically required for operating cost parity. Octopus Energy, an energy retailer and

heat pump installer in the UK, aims for a SCOP of ~3.3 on its installations.

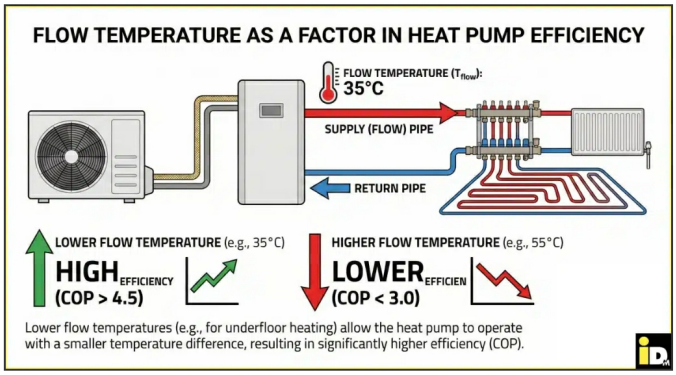
EXHIBIT 9: **Illustrative comparative economics of a gas boiler vs heat pump at varying efficiency levels and power price scenarios**

£ except where stated	Gas combi boiler	UK heat pump pre grant	UK heat pump post grant	France illustrative heat pump post grant
Upfront cost	3,500	10,000	2,500	2,241
Asset Life (years)	13	20	20	20
Depreciation and financing	406	872	218	195
Annual maintenance	110	180	180	180
Annual fuel costs (at 300% COP for HP)	946	1,044	1,044	776
Total cost of ownership (at 300% COP for HP)	**1,462**	**2,096**	**1,442**	**1,151**
Annual saving vs gas boiler (at 300% COP for HP)	0	-634	19	808
Annual fuel costs (at 400% COP for HP)	946	783	783	582
Total cost of ownership (at 400% COP for HP)	**1,462**	**1,835**	**1,181**	**957**
Annual saving vs gas boiler (at 400% COP for HP)	0	-373	280	1,002

Source: Bernstein analysis and estimates

Alongside the quality of installation, a key driver of achieved heat pump operating efficiency is the level of insulation in a property; the better the insulation, the lower the loss of heat from the property and the lower the required "flow temperature" from the heat pump, where the flow temperature is the temperature of the water in the system as it leaves the pump unit. A better-insulated property can operate at a lower flow temperature because the property's radiators (or other heat emitters) will need to work less hard to maintain the targeted temperature given the reduced heat leakage. This reduces the burden on the heat pump and increases its efficiency.

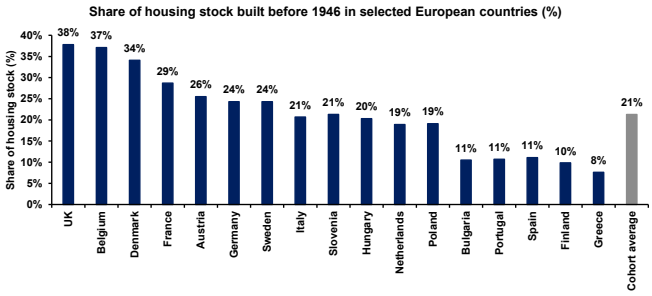
EXHIBIT 10: **Flow temperatures, impacted by insulation levels, are a key driver of heat pump efficiency**



Source: IDM Energie

We set out an overview of the comparative ages of the housing stock in major European countries in Exhibit 11. The age of housing stock is strongly correlated with insulation quality.

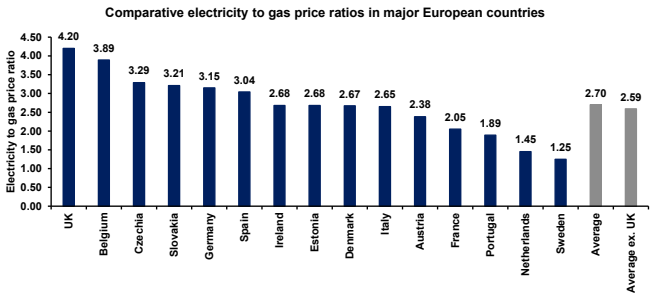
EXHIBIT 11: **Comparative analysis of the share of housing stock in major European countries that was built before 1946**



Source: Building Research Establishment, Bernstein analysis

Alongside the importance of operating efficiency, Exhibit 9 above highlights the importance of the electricity-to-gas price ratio in driving the competitiveness of heat pumps. We set out a comparison of the electricity-to-gas price ratios of major European countries in Exhibit 12. As we explore in more detail later in this Blast, there is a strong correlation between this metric and heat pump uptake in major European countries.

EXHIBIT 12: **The UK and Belgium have the highest electricity-to-gas price ratios amongst major European countries**



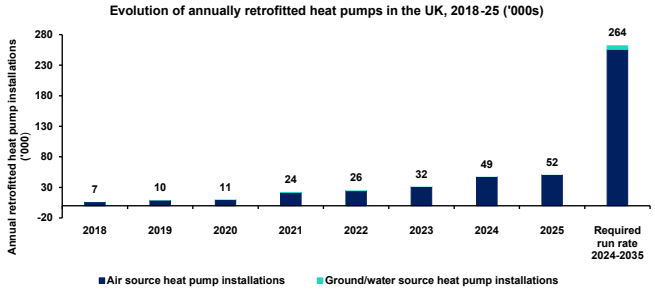
Source: Nesta, Bernstein analysis

HEAT PUMPS: A MIXED ADOPTION PICTURE IN EUROPE

As set out in Exhibit 13, heat pumps are currently a niche technology in markets such as the UK, with annual installations currently tracking far short of the minimum contemplated annual run rate in the UK's electricity system planning (~250,000 year to 2035) or in UK government targets (450,000/year by 2030). Regulator Ofgem has highlighted the challenge well : "Earlier this year, the UK government updated its ambition for heat pump uptake in the Warm Homes Plan (WHP), replacing the previous aspiration of 600,000 installations per year by 2028 with an ambition for 450,000 annual installations by 2030. Official statistics show there were 51,886 retrofit heat pump installations in 2025.13 If around 200,000 installations per year in 2030 are expected to come from new homes, retrofit installations in existing buildings would need to rise to around 250,000 per year by 2030, implying average growth

of around 37% per year over the next five years."

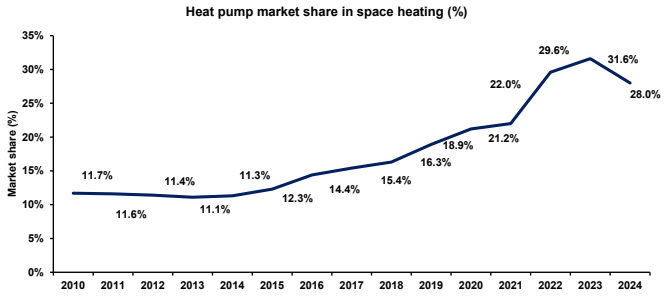
EXHIBIT 13: **Annual heat pump installations in the UK are currently tracking far below the run rate contemplated in electricity system planning**



Source: DESNZ, National Energy System Operator, Bernstein analysis

In Europe more broadly, as we show in Exhibit 14, the picture is different, with heat pumps now a mainstream heating technology, reaching ~30% market share in 2022, 2023 and 2024 (the most recent year for which data is available). The UK's laggard status in heat pump adoption is most likely explained by a combination of factors including 1) the country's high electricity-to-power price costs as discussed in more detail earlier and 2) the comparatively poor levels of insulation in UK real estate, which serves to increase heat pump annual running costs.

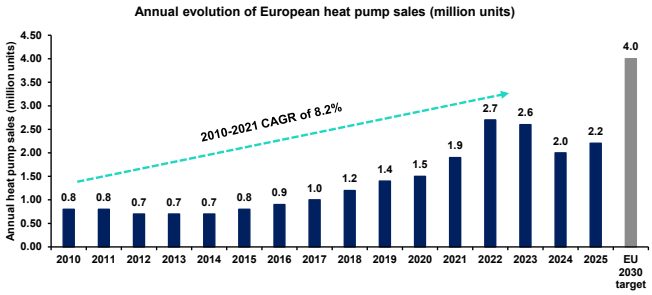
EXHIBIT 14: **Annual market share of heat pumps in European space heating, 2010-2024**



Source: European Heat Pump Association, Bernstein analysis

We set out the annual evolution of heat pump sales in Europe in Exhibit 15. Following an ~8% CAGR from 2010-2021, sales spiked by 42% between 2021 and 2022 in the wake of the Russia-Ukraine crisis, before falling by ~25% between 2022 and 2024, and rising by ~10% between 2024 and 2025. Annual sales remain significantly short of the EU's 2030 target of 4m units. For a cohort of European countries, heat-pump sales in Q1'26 have risen by 17%, as discussed below.

EXHIBIT 15: **Annual evolution of European heat pump sales**



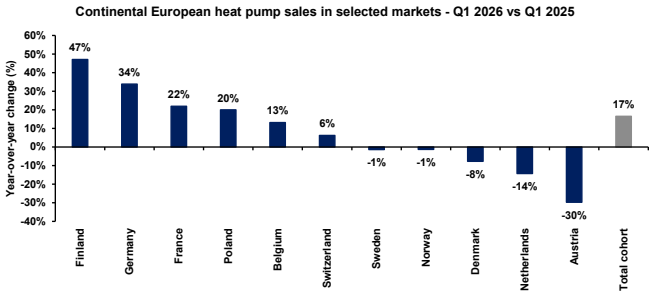
Source: European Heat Pump Association, Bernstein analysis

ON THE BRINK OF A BREAKTHROUGH...?

There is significant evidence that the current Iranian crisis is catalyzing decarbonization momentum in Europe, as seen e.g. in 1) rising EV sales and 2) increasing installation of solar panels. Some industry observers have suggested that heat pump installation could see a similar boost. In the UK, *The Times* reported on 30 April 2026 that "the number of applications for heat pump grants...jumped to its second-highest monthly level on record" in March 2026. Similarly, in France, *Le Monde* reported in May 2026 that environmental services provider Effy had seen a >140% increase in heat pump installation inquiries in April 2026 vs the prior year. The French association of heating, ventilation, air conditioning and refrigeration industries (UNICLIMA) has also reported a "boom" in requests for quotes.

As we set out in Exhibit 16, some major continental European markets saw strong year-over-year growth in heat pump sales in Q1 2026, with Finland reporting a 47% increase, Germany a 34% increase and France a 22% increase.

EXHIBIT 16: **Q1 2026 year-over-year heat pump sales data in selected continental European markets**



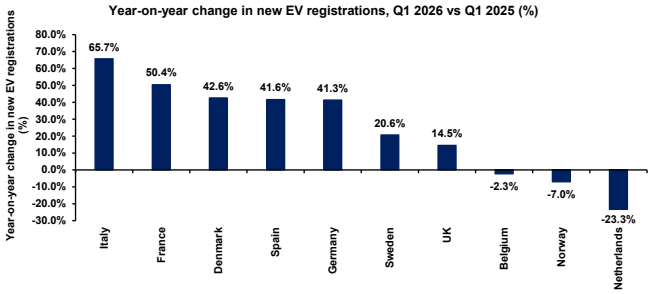
Source: European Heat Pump Association, Bernstein analysis

OTHER DECARBONISATION TECHNOLOGIES ARE SEEING A BOOST FROM THE IRANIAN CRISIS

The UK is currently seeing significant momentum in EV adoption, with the Society of Motor Manufacturers and Traders reporting a 59.1% year-on-year increase in Battery Electric Vehicle

registrations in April 2026, equivalent to a ~26% market share. As set out in Exhibit 17, in Europe more broadly, Italy saw a ~66% year-on-year increase in new battery electric vehicle registrations in Q1 2026 vs 2025.

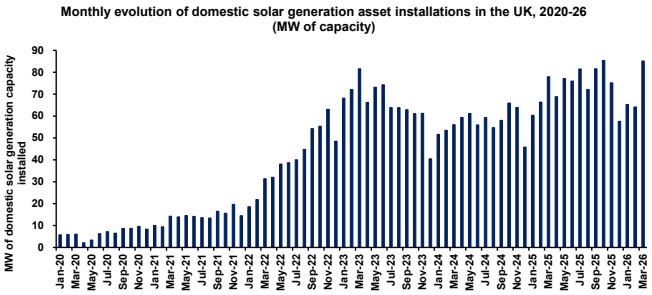
EXHIBIT 17: **Europe saw significant increases in EV demand in Q1 2026 vs Q1 2025**



Source: ACEA, Bernstein analysis

As set out in Exhibit 18, a similar trend is visible in domestic solar generation capacity in the UK. 85MW of capacity was installed in March 2026, c.4% above the 2023 peak in March (~82GW) and close to the maximum monthly figure since the start of 2020 (85MW in April 2025). Since the start of the Russia-Ukraine war in February 2022, monthly domestic solar panel capacity installation has risen by ~300% in the UK.

EXHIBIT 18: **UK monthly domestic solar generation capacity installations**



Source: DESNZ, Bernstein analysis

In support of the view that heat pumps would be also caught up in the post-war decarbonisation boom, as set out in Exhibit 19, *The Times* in the UK reported on 30 April 2026 that in March 2026, the number of applications for heat pump grants jumped to its second-highest level on record. Meanwhile Greg Jackson, CEO of Octopus Energy, told the BBC in March 2026 that his company had seen a 50% rise in solar panels vs February 2026, a 30% rise in heat pump sales and a >33% rise in enquiries about EVs.

EXHIBIT 19: **The 2026 Iranian crisis has spurred adoption of decarbonisation technologies in the UK**

Solar panels fitted every three minutes in UK since Iran conflict

More of them fitted in March than any month since 2012 as conflict in the Middle East causes energy crisis

Solar panels can help households save hundreds of pounds a year

ADRIAN SHERRATT FOR THE TIMES

Adam Vaughan, Environment Editor
Thursday April 30 2026, 8.45pm, The Times

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The energy crisis triggered by conflict in the Middle East has caused a rush in solar panel and heat pump installations, with solar power being fitted on a rooftop about every three minutes.

There were more than 27,000 solar power installations in March, the highest monthly total for 14 years. The number of applications for heat pump grants also jumped to its second highest monthly level on record.

Source: The Times

INCREASING INNOVATION COULD DRIVE COSTS LOWER

In a 2023 report for the UK Energy Research Centre, Phil Heptonstall and Mark Winskel found that "in the UK, there has been little or no reduction in the average total installed cost of heat pumps over the past decade. Over the same period, some cost reductions have been achieved internationally, and over time, particular countries have successfully aligned market growth with reduced installed costs. There are real prospects for reduced installed costs, particularly non-equipment costs, over the next decade. Most UK forecasts suggest a reduction in total installed costs by 2030 of around 20-25%. Expected savings are higher for non-equipment installations costs, including labor costs, than for equipment costs."

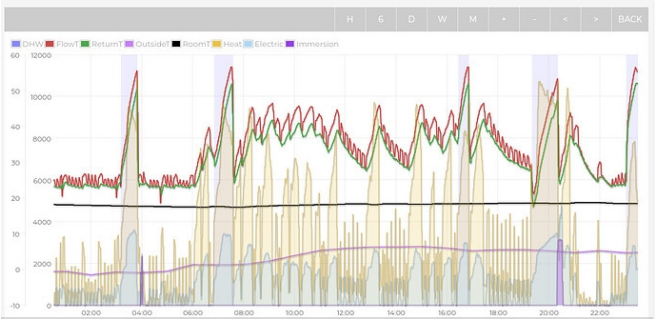
Operating efficiency is showing a favourable trend

There is some evidence that the operating efficiency of heat pumps could meaningfully improve over time. Jan Rosenow, Professor of Energy and Climate Policy at Oxford University, in partnership with Trystan Lea, analyzed a cohort of heat pumps installed on

heatpumpmonitor.org, finding an average Seasonal Coefficient of Performance of ~3.9 for this cohort, vs 2.81 in the cohort used for the UK government's Electrification of Heat trial in December 2024. The authors estimated that these results were equivalent to a 19% annual operating cost premium to gas for the UK government cohort vs a 17% saving for the Heat Pump monitor cohort. As installer expertise and experience increase over time, we see potential for accompanying improvements in operating efficiency and cost competitiveness.

We set out in Exhibit 20 the operating data for a poorly-performing heat pump from the UK government's Electrification of Heat trial program. The data, for system EOH0567 on the Heat Pump Monitor website, shows poor temperature control and short cycling, indicating sub-optimal energy efficiency. With installer knowledge and experience growing as the technology is more widely deployed, achieved efficiency levels should improve.

EXHIBIT 20: **Operating data for a poorly-performing heat pump from the UK government's Electrification of Heat trial program**



Source: Open Energy Monitor

There is some room for optimism on installation costs

In contrast to the more pessimistic picture painted above, the UK government estimated in March 2026 that there had been an 11% real terms reduction in installation costs for heat pumps in the UK since 2022. Additionally, there has also been significant innovation in the installation industry to address specific adoption bottlenecks. For example, as the UK government explains, "to provide hot water, heat pumps require a water storage cylinder, similar to older generations of gas boiler systems. New combi gas boilers do not require these cylinders, so approximately 50% of the UK housing stock lack water storage cylinders now. Further, smaller homes and last lack the spare storage space to install cylinders." To address this bottleneck, some installers have devised smaller hot water cylinders that can fit into existing kitchen cabinetry. We see potential for further innovation to drive installation costs lower.

Manufacturing process improvements are also a likely affordability tailwind

We also see potential for the unit costs of heat pumps to decline as manufacturing ramps up, with companies such as Octopus Energy in the UK currently making bold moves to establish a leading position in the technology.

Octopus Energy manufacture heat pumps in a factory in Northern Ireland, with full integration of deployment across the initial design, the manufacturing process and final installation and commissioning. Octopus's heat pumps are integrated into the Octopus Energy customer app and are fitted with sensors to enable remote performance monitoring.

EXHIBIT 21: **Octopus Energy manufacture heat pumps under their Cosy brand name**

Source: The Daily Telegraph

The IEA explain that "heat pump production costs vary widely by region, shaped by general manufacturing conditions (notably skills, infrastructure and labor) and market-specific factors such as product type (typically air-to-air or air-to-water), unit size, compressor choice, performance standards and regulations. Final assembly requires little energy or raw materials, so components dominate overall costs, accounting for 60-85% of production costs. Compressors are the most complex part, representing about 30% of production costs alone."

THERE IS A ROBUST OUTLOOK FOR EUROPEAN GOVERNMENT SUPPORT FOR HEAT PUMPS

Whilst heat pump competitiveness vs gas boilers is dependent on government subsidy, as we set out in more detail below and as shown in Exhibit 22, we see a robust appetite from governments in Europe to support the technology. The April 2026 AccelerateEU communication sets out that "the Commission will promote and help develop, including through the Energy Transition Investment Council and the Energy Efficiency Financing Coalition, social leasing schemes for clean and efficient technologies which Member States are encouraged to use to support the rapid uptake of, e.g. e-vehicles, residential heat pumps and small-scale batteries. The Commission stands ready to assist Member States in establishing financial incentives, such as targeted tax credits, for the rapid deployment of clean energy technologies such as electric

vehicles, industrial and household heat pumps, behind-the-meter batteries and industrial thermal storage, while ensuring compliance with commitments under fiscal rules."

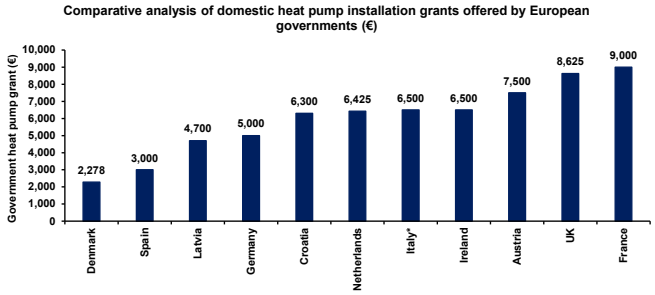
EXHIBIT 22: **Comparative heat pump installation costs in selected major European countries**

Country	Price range (pre-tax)	Explanatory factors
Poland	€9,000-€15,000	Lower labour costs, reduced bureaucracy
UK	€10,000-€14,000	0% VAT, simpler regulatory standards
Netherlands	€11,000-€19,000	Large variety, standardised systems
France	€12,000-€20,000	Fixed funding, medium price level
Austria	€13,500-€20,000	Similar price level to Germany but with VAT concessions
Germany	€23,000-€40,000	Highest price level in the peer group

Source: Heat Pumps Watch, Bernstein analysis

We set out an overview of the subsidies available for heat pump installation in selected European countries in Exhibit 23.

EXHIBIT 23: **Many European governments offer generous subsidies for heat pump adoption**

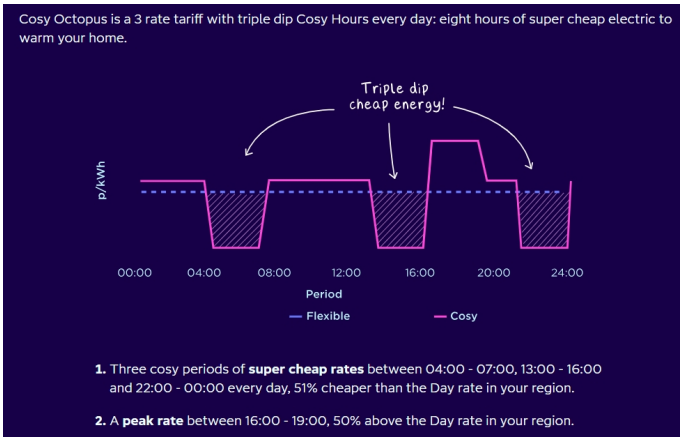


Source: DESNZ, European Heat Pump Association, Bernstein analysis

INNOVATIVE TARIFFS CAN IMPROVE COST COMPETITIVENESS

Alongside innovation in the design and manufacturing process, Octopus are also enabling the heat pump roll-out with innovative tariff options, including the Octopus Cosy tariff (see Exhibit 24), under which customers are offered 8 hours of cheap electricity per day, split over two 2-3 hour slots. During these discounted periods, customers are charged ~14.8p per KWh vs 52.49p/kWh outside of these times. Given the very high peak rate under the Cosy tariff, we would not expect it to appeal to customers who don't own a battery.

EXHIBIT 24: **Octopus Energy's Cosy tariff is targeted at heat pump owners and offers 3 separated discounted electricity blocks**



Source: Octopus Energy

OR ANOTHER FALSE START?

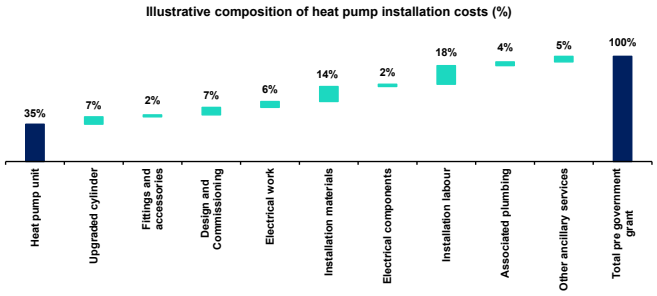
Despite the potential for optimism surrounding heat pumps in countries such as the UK, we would caution that significant barriers remain against widespread adoption as a mainstream domestic heating technology, principally the high installation costs and the UK's high electricity-to-gas price ratio. In this section we set out these potential obstacles in more detail.

COST COMPETITIVENESS REMAINS RELIANT ON GOVERNMENT SUBSIDY

Heat pumps are considerably more expensive to install than the equivalent gas boiler. EDF present indicative cost ranges for an air source heat pump and a comparable gas boiler as between £3,999-£10,000 and £3,000-£4,000 respectively, with the stated air source heat pump range incorporating the impact of the government grant. Additionally, the air source heat pump range assumes no required upgrades to either the existing hot water cylinder or radiator system. Given that these ugrades will be required in many instances, it is clear that the upfront cost of a heat pump remains significantly higher than an equivalent gas boiler even including government grants, and is unviable without subsidy.

We set out an illustrative decomposition of UK heat pump installation costs in Exhibit 25 below.

EXHIBIT 25: **Illustrative decomposition of UK heat pump installation costs**



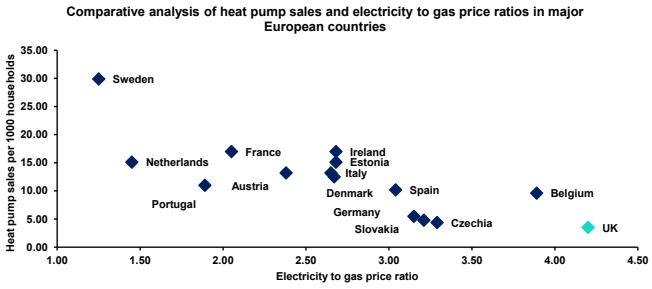
Source: Bernstein analysis and estimates

In relation to market dependence on government subsidy, we would further note key trends set out in Exhibit 16 above. In Austria, where there is no government subsidy for heat pump installation, heat pump sales fell ~30% year-over-year in Q1 2026. Additionally, European Heat Pump Association commentary attributed the strength in German heat pump installations in Q1 2026 to attempts to preempt subsidy reform. In Sweden, where subsidies were lowered, heat pump sales fell ~1% year-on-year in Q1 2026.

MORE SUSTAINED UPTAKE IN THE EUROPEAN LAGGARDS WILL LIKELY REQUIRE A DECREASE IN THE ELECTRICITY TO GAS PRICE RATIO

As set out in Exhibit 26, countries such as the UK and Belgium significantly lag other European countries in heat pump adoption, with a clear correlation to their higher electricity-to-gas price ratios. As previously noted, this is a significant driver of operating cost competitiveness vs alternative heating technologies such as gas boilers; a higher electricity-to-gas price ratio reduces or even completely erodes the economic case for heat pumps, with even the most efficient installations only neutralising the fuel cost disadvantage vs alternative technologies such as gas boilers, and the least efficient installations often falling short altogether. The most efficient installations may also incur higher upfront costs to optimize the asset, again weakening the overall economic case. As such, it is clear that in order to make meaningful headway in heat pump adoption, countries such as the UK must reduce their electricity-to-gas price ratios.

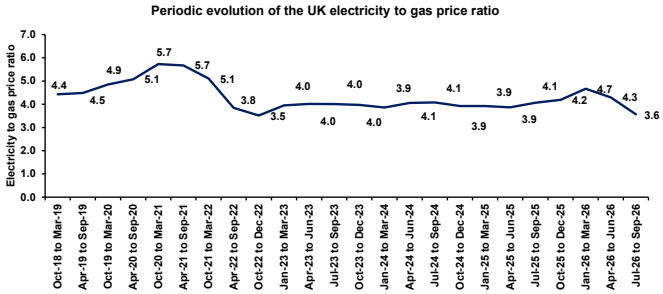
EXHIBIT 26: **The UK's limited heat pump adoption so far is substantially explained by its high electricity-to-gas price ratio**



Source: Nesta, European Heat Pump Association, Eurostat, Bernstein analysis

As set out in Exhibit 27, the UK's most recent price cap announcement outlined tariffs consistent with a slight fall in the electricity-to-gas price ratio, with the gas unit cost moving from 5.74p/kWh to 7.33p/kWh and the electricity unit cost moving from 24.67p/kWh to 26.11p/kWh.

EXHIBIT 27: **The UK's most recent electricity price cap update showed a decline in the electricity-to-gas price ratio**



Source: Nesta, Ofgem, Bernstein analysis

WHAT ARE THE SYSTEM IMPLICATIONS OF HEAT PUMP ACCELERATION?

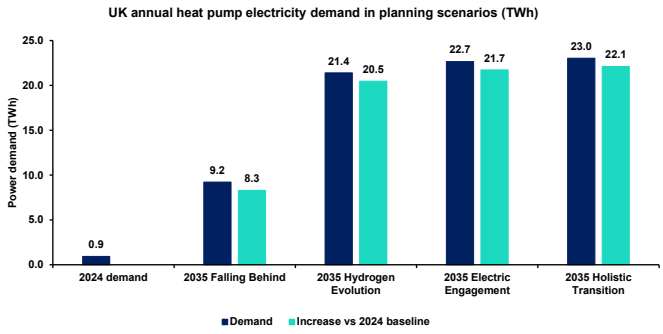
In the event that if countries are successful in spurring widespread adoption of heat pump technology, we see far-reaching consequences for the electricity system as a whole, necessitating further distribution grid upgrades and materially increasing overall demand.

WIDER HEAT PUMP ADOPTION WOULD BE A SIGNIFICANT TAILWIND FOR TOTAL CONSUMPTION AND PEAK DEMAND...

The average heat pump in the UK could be expected to use ~2-4MWh of electricity per year, implying potentially significant upside to UK power demand in the event of a sustained increase in adoption. Under the UK National Energy System Operator's planning assumptions, it is contemplated that by 2035 between

~3m and ~8m households will use heat pumps, vs ~300,000 in 2024. As set out in Exhibit 28, this would imply between 8 and 22 TWh of additional electricity demand from heat pumps by 2035, equivalent to 5-14% of 2024 total demand.

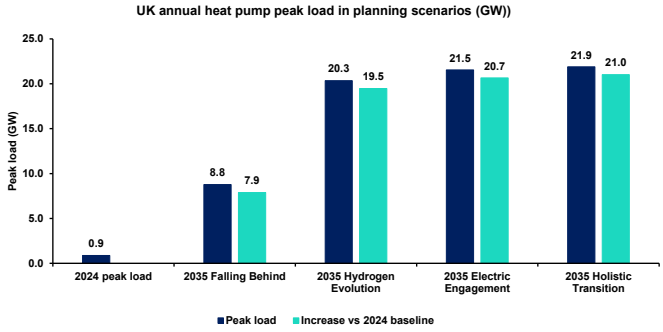
EXHIBIT 28: **The UK's electricity system planning assumptions imply 17-45TWh of additional electricity demand from heat pumps by 2035**



Source: National Energy System Operator, Bernstein analysis

As set out in Exhibit 29, the potential impact is still more significant when analysed in terms of peak load. The planning scenarios imply a peak load increase from heat pumps of 8-21GW by 2035, equivalent to 17-47% of peak load in 2024. This underlines the potential network reinforcement need that wider heat pump adoption could drive.

EXHIBIT 29: **The UK's electricity system planning assumptions imply a peak load from heat pumps 8-21GW higher by 2035**



Source: National Energy System Operator, Bernstein analysis

...DRIVING GROWTH FOR ELECTRICITY DISTRIBUTION NETWORKS AND INCREASED OPPORTUNITY FOR GENERATORS

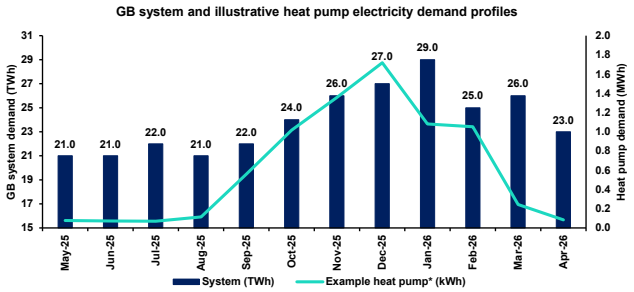
Given the significant increase in peak load that heat pumps could drive, we see electricity distribution companies as the likely primary beneficiaries, with growth driven by the need for network reinforcement to handle the increased load. Additionally, we would see opportunities for renewable and thermal generators given the likely increase in overall demand.

Distribution networks

Increasing roll-out of heat pumps will necessitate significant investment to reinforce the electricity distribution grid. A 2017 study in *Applied Energy* explains that "dwellings in areas that are connected to the gas network will generally have distribution network capacities designed for very little electric heating. The problems associated with connecting large numbers of heat pumps are excessive voltage drop beyond allowed limits and insufficient thermal capacity of the Low Voltage feeder and transformer leading to overheating of these elements unless they are reinforced." The electricity demand is impact is exacerbated by seasonality; the same study reports that "during winter periods heating energy demand can reach around 5 times the magnitude of electricity demand in UK dwellings." The potential demand impact is further exacerbated by the fact that heat pump efficiency is at its lowest in the cold temperatures in which usage is highest.

As we set out in Exhibit 30, the monthly electricity usage profile of a heat pump in the UK is closely aligned with that of the broader system, increasing the system impact of the heat pump roll out.

EXHIBIT 30: **GB system and illustrative heat pump monthly demand profiles**



*Example heat pump is system 175 from Heat Pump Monitor
Source: NESO, Heat Pump Monitor, Bernstein analysis

Other system impacts

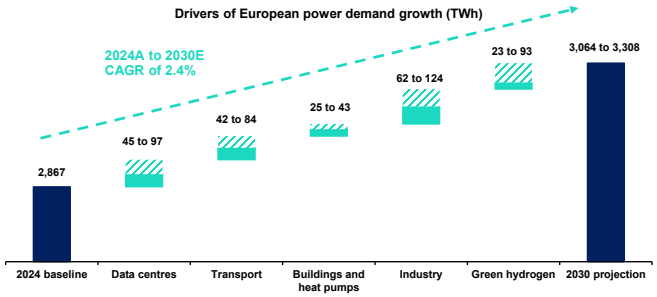
We see other significant potential impacts on total power consumption and potentially system flexibility.

Demand

Heat pumps have the potential to significantly increase electricity demand in Europe. Taking the UK as an example, a 2023 study in *Energy and Buildings* developed a statistical model to "predict the additional load on the GB electricity generation and distribution infrastructure for various current and future (2050) climates, dwelling energy efficiencies and heat pump deployment scenarios", finding that "for a cold year in the 2020s, a 100% uptake of heat pumps in the existing GB dwelling stock gave a peak electricity demand for the heat pumps of 78GW and an annual electricity demand of 189 TWh...this represents an increase in the GB peak

electricity demand in excess of 100% and an annual electricity demand increase of around 60%."

EXHIBIT 31: **Heat pumps sit alongside other drivers that in aggregate could push European power demand materially higher by 2030**

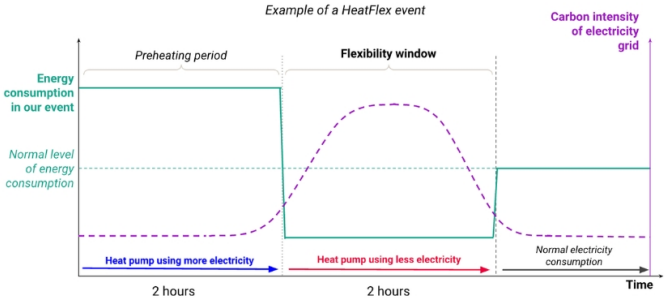


Source: McKinsey, BNEF, IEA, Hydrogen Europe, Eurelectric, European Heat Pump Association, Bernstein analysis and estimates

Flexibility

Heat pumps offer a potential future electricity system stability contribution in the form of demand side response or demand flexibility. Where the system is approaching a demand peak, heat pumps participating in a flexibility services program can be turned off for a short period, easing the demand strain on the grid with little impact on user experience given that buildings retain heat for some time after a heat source has been turned off. The research and innovation foundation Nesta ran a trial between February and April 2024 where they remotely controlled participants' heat pumps for 4 hours during 'HeatFlex' events, establishing the viability of a demand flexibility program - see Exhibit 32.

EXHIBIT 32: **Nesta HeatFlex illustration**



Source: Nesta

CONCLUSIONS

Clearly heat pump adoption will be a significant boost to electrification of the economy and the present crisis is clearly boosting prospects once again. However, at present the technology still requires a combination of subsidies, high insulation levels and a favorable electricity-to-gas price ratio to be viable. If heat pump costs can follow the path of batteries (Battery Energy Storage Systems - A primer for investors on their economics) and solar, the technology will see more widespread adoption, increasing power demand as well as requiring more investments in electricity distribution grids. We support UK regulator's more cautious approach on not investing ahead of heat pump demand in the upcoming electricity distribution period.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	3 Jun 2026 Closing Price	Price Target	TTM Rel. Perf.	Adjusted EPS Cur	Adjusted EPS 2025A	Adjusted EPS 2026E	Adjusted EPS 2027E	Adjusted P/E (x) 2025A	Adjusted P/E (x) 2026E	Adjusted P/E (x) 2027E
EOAN.GR (E.ON)	M	EUR	18.09	18.00	3.7%	EUR	1.16	1.07	1.22	15.6	16.9	14.8
FORTUM.FH (Fortum)	U	EUR	20.84	17.70	20.4%	EUR	0.82	1.09	0.98	25.3	19.1	21.2
NG/.LN (National Grid)	O	GBp	1,199.00	1,450.00	2.0%	GBP	0.73	0.77	0.89	16.4	15.5	13.5
ORSTED.DC (Orsted)	M	DKK	160.35	140.00	(5.1)%	DKK	1.95	7.61	7.84	82.3	21.1	20.5
RWE.GR (RWE)	M	EUR	56.86	57.00	54.2%	EUR	2.48	2.57	3.02	22.9	22.1	18.8
SSE.LN (SSE)	O	GBp	2,388.00	3,040.00	23.9%	GBP	1.61	1.51	1.80	14.9	15.8	13.3
VWS.DC (Vestas Wind)	O	DKK	173.40	195.00	50.1%	EUR	0.78	1.14	1.36	29.9	20.3	17.1
EDME			1,548.21									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

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